Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)		
Lab exp 4	Date:	Enrolment No:92400133079

CODE

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import butter, bilinear, freqz, cheby1


# Design Butterworth Filter
def design_butterworth_filter(filter_order, cutoff_frequency, sampling_frequency):
    analog_b, analog_a = butter(
        filter_order,
        cutoff_frequency,
        analog=True,
        btype='low'
    )

    # Perform bilinear transformation
    digital_b, digital_a = bilinear(
        analog_b,
        analog_a,
        sampling_frequency
    )

    return digital_b, digital_a


# Design Chebyshev Filter
def design_chebyshev_filter(filter_order, cutoff_frequency, sampling_frequency,
ripple):
    analog_b, analog_a = cheby1(
        filter_order,
        ripple,
        cutoff_frequency,
        analog=True,
        btype='low'
    )
```

```
    # Perform bilinear transformation
    digital_b, digital_a = bilinear(
        analog_b,
        analog_a,
        sampling_frequency
    )

    return digital_b, digital_a


# Plot Filter Response
def plot_filter_response(digital_b, digital_a, sampling_frequency):

    # Compute frequency response
    frequency, magnitude_response = freqz(
        digital_b,
        digital_a,
        fs=sampling_frequency
    )

    # Plot magnitude response
    plt.figure(figsize=(10, 6))
    plt.plot(frequency, np.abs(magnitude_response))
    plt.title('Filter Magnitude Response')
    plt.xlabel('Frequency (Hz)')
    plt.ylabel('Magnitude')
    plt.grid(True)
    plt.show()

    # Compute response for plotting
    _, impulse_response = freqz(
        digital_b,
        digital_a,
        worN=4096
    )

    # Plot response
    plt.figure(figsize=(10, 6))
```

```
    plt.plot(np.real(impulse_response))
    plt.title('Filter Impulse Response')
    plt.xlabel('Samples')
    plt.ylabel('Amplitude')
    plt.grid(True)
    plt.show()


# Filter Specifications
filter_order = 4
cutoff_frequency = 1000
sampling_frequency = 8000
ripple = 0.5

# Design and plot Butterworth filter
digital_b, digital_a = design_butterworth_filter(
    filter_order,
    cutoff_frequency,
    sampling_frequency
)

plot_filter_response(
    digital_b,
    digital_a,
    sampling_frequency
)

# Design and plot Chebyshev filter
digital_b, digital_a = design_chebyshev_filter(
    filter_order,
    cutoff_frequency,
    sampling_frequency,
    ripple
)

plot_filter_response(
    digital_b,
    digital_a,
    sampling_frequency
```

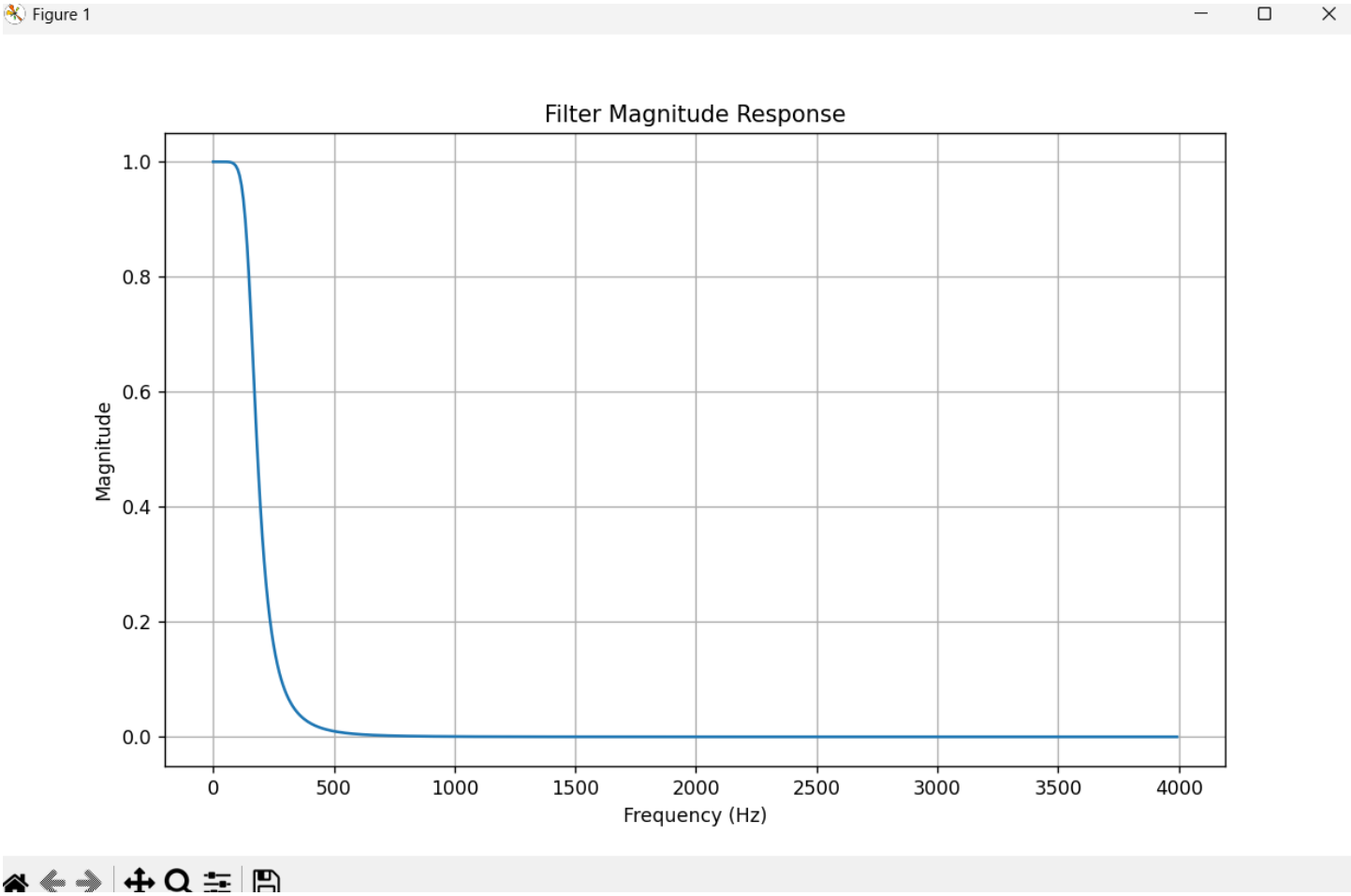
```
)

# Save filter coefficients
filter_path = 'filter_coefficients.txt'

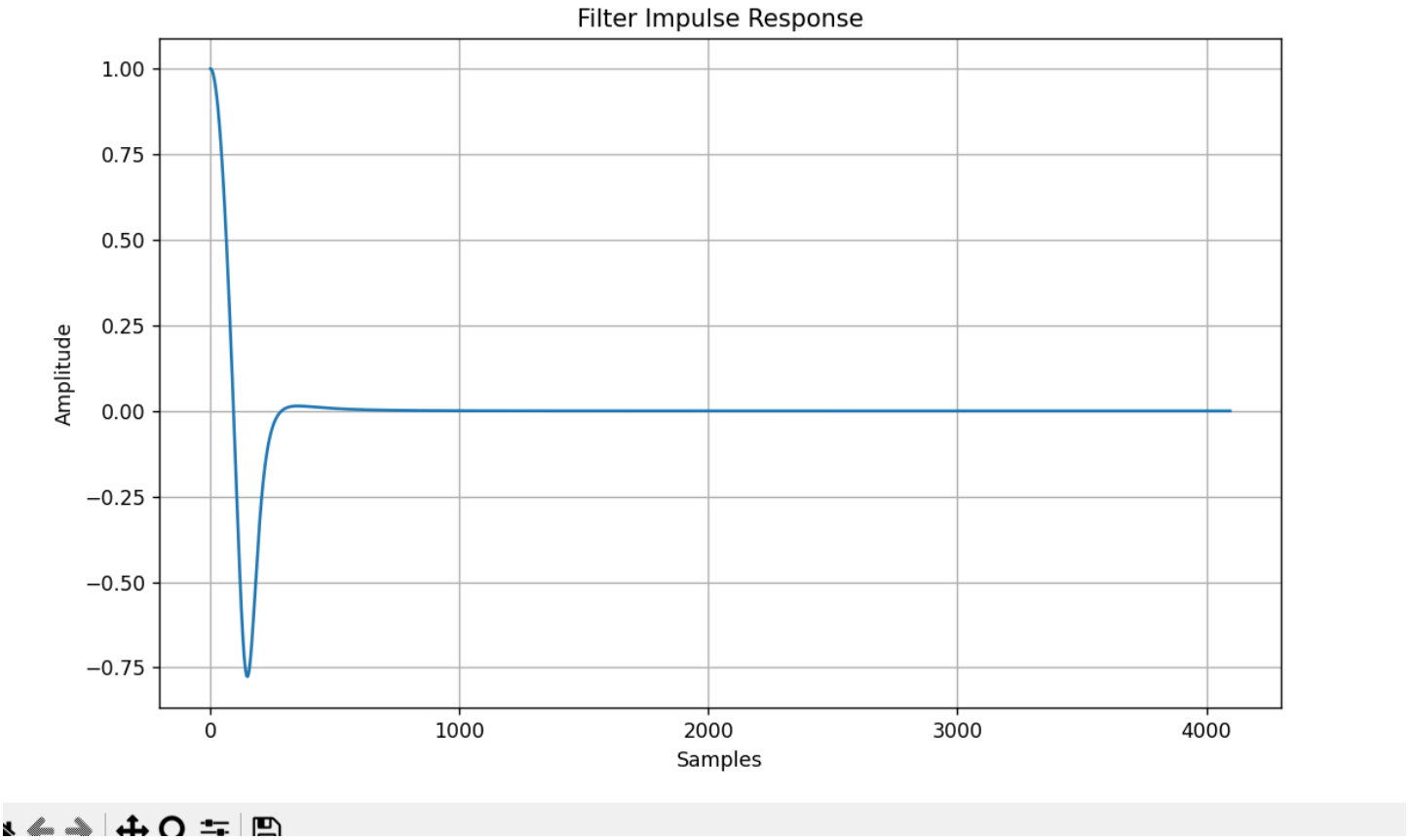
np.savetxt(
    filter_path,
    np.vstack((digital_b, digital_a)),
    delimiter=','
)

print(f"Filter coefficients saved at: {filter_path}")
```

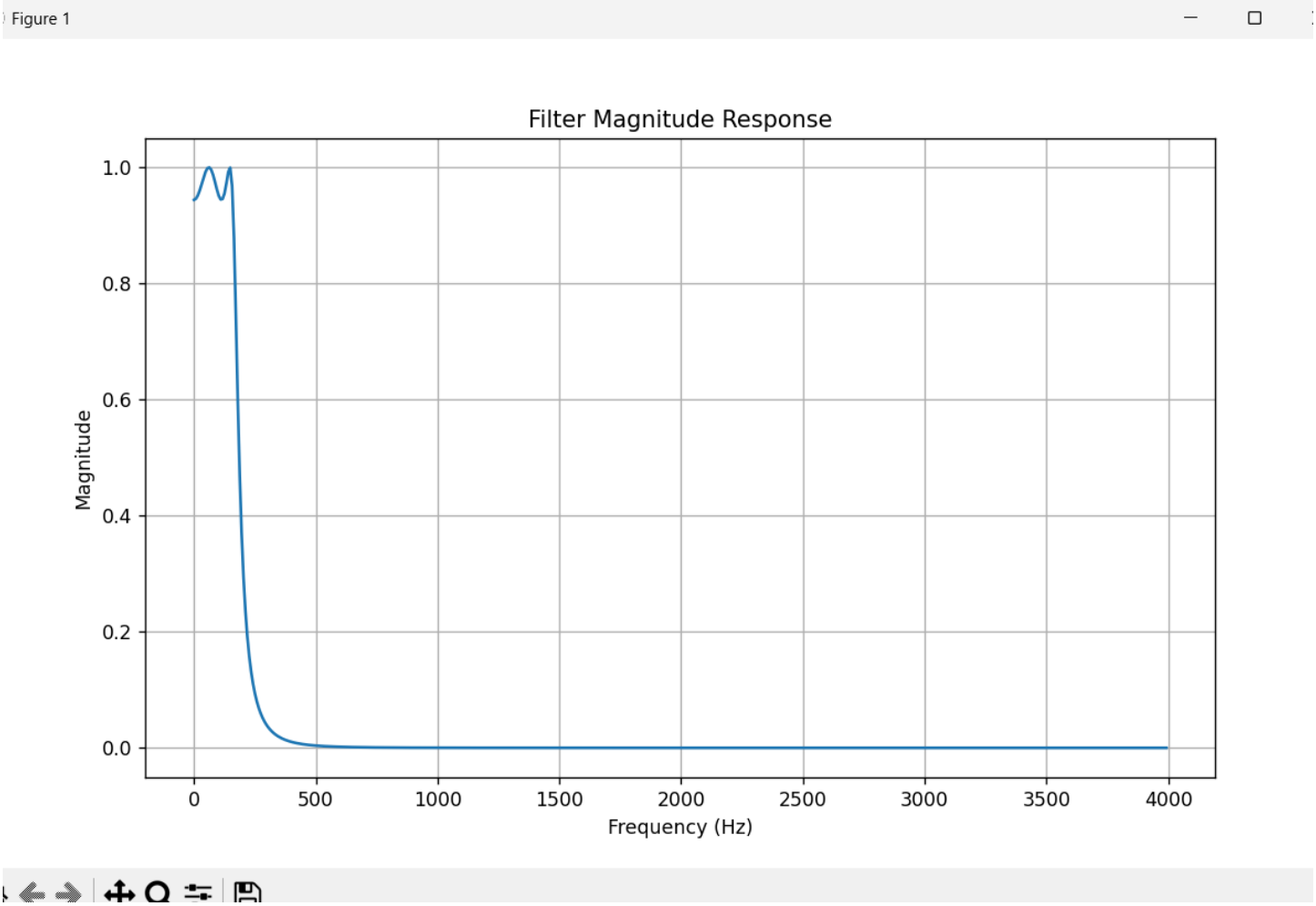
OUTPUT



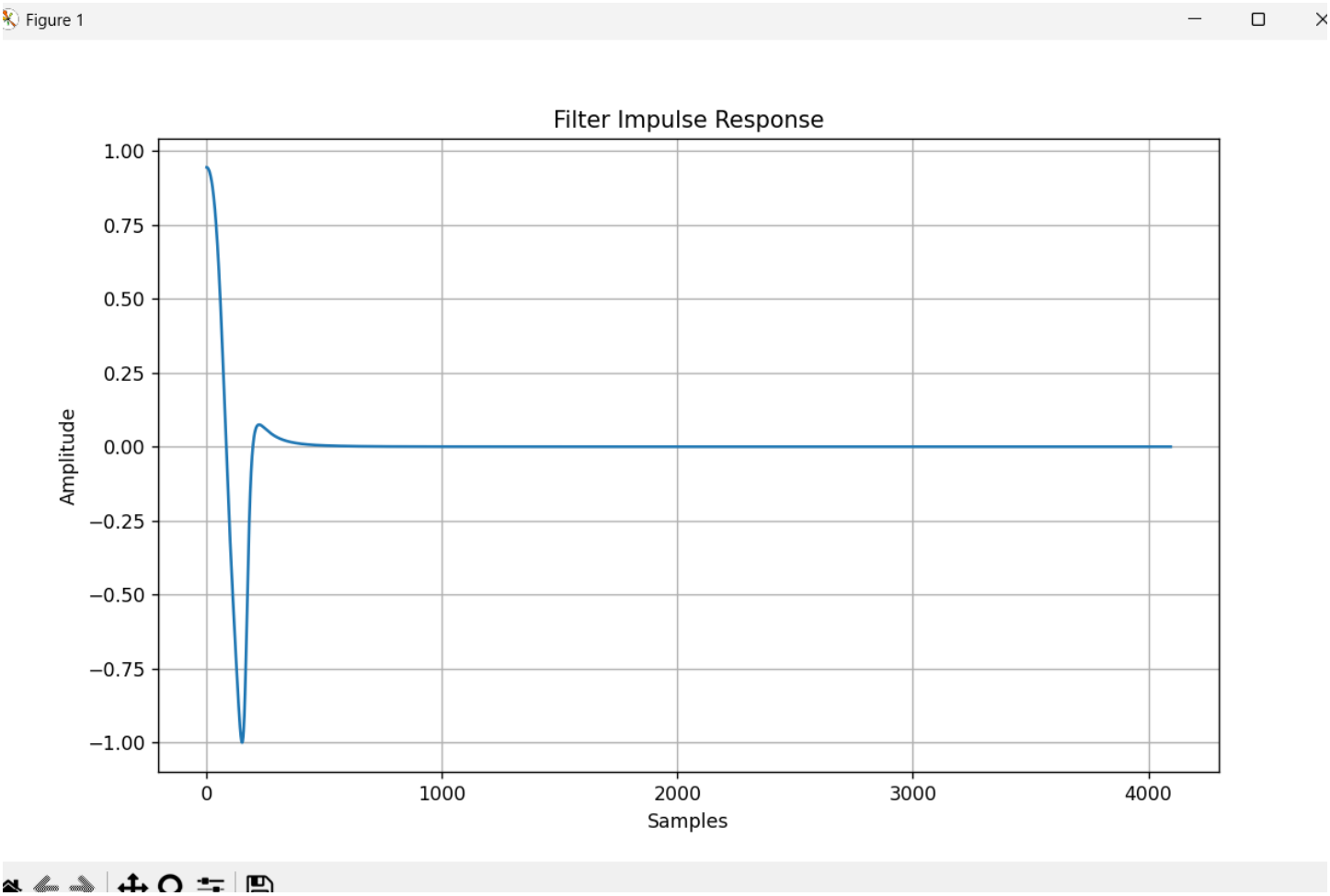
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GITHUB : https://github.com/yugmakwana99/DSIP.git